Errata for *Adaptive Experimental Design: Prospects and Applications in Political Science*

The maintained replication code behind these errata is at https://github.com/active-maintenance-aec/offer-westort_coppock_green_2021 and is archived at Zenodo at https://doi.org/10.5281/zenodo.24168408, a DOI that always resolves to the latest version.

Fifteen corrections to Offer-Westort, Coppock, and Green (2021), *American Journal of Political Science* 65(4): 826–844, doi:10.1111/ajps.12597. Four are sentences in the article stating a number or a treatment label that the study's own data do not support, three are reference-list entries, and eight are in the supporting information. No conclusion in either document changes, no result in either is wrong, and every figure and table is correct as printed.

The corrected values in entries 1 to 3 are computed from a maintained reproduction of the analysis, held at `active-maintenance-aec/offer-westort_coppock_green_2021`, which reproduces all 913 of the article's published table and figure values; the reference-list entries are verified against the authoritative record at Crossref, named in each entry.

The supporting information entries come from a reading of sections B.2, B.3, C.1, C.2 and C.3 against the authors' own source, whose compiled PDF is byte-identical to the published appendix. They are of a different kind from the article's seven, because none of them changes a quantity: every numeric claim in those five sections was recomputed independently and all of them reproduce, including both columns of all twelve rows of Table C.1. What is left is notation, one word, two dropped or transposed characters, one variance term, and one number that is right and is asked to demonstrate something it does not. Their page numbers are the appendix's own, which is separately paginated 1 to 49.

Changed tokens are shown in bold on both sides, and a changed expression is set in bold mathematics, which is the same mark made the way mathematics can carry it. Where a sentence contains other quantities, those are reprinted unchanged. one entry, 8, quotes a passage and proposes no change to it; its note says why.

1. The estimated mean of the leading minimum wage arm

> The winning arm, by a hair, was Proposal 3 (**B**, without current minimum wage), with a probability of being best of 0.219 and an estimated mean of **0.895** over 183 respondents. (p. 835)

Corrected: The winning arm, by a hair, was Proposal 3 (**N**, without current minimum wage), with a probability of being best of 0.219 and an estimated mean of **0.865** over 183 respondents.

The arm with a 0.219 probability of being best and 183 respondents is Proposal 3 (N), whose inverse-probability-weighted mean is 0.865. The 0.895 printed in the sentence is Figure 4's estimate for Proposal 3 (Y), a different arm, which drew 121 respondents and ended with a 0.120 probability of being best. Figures 3 and 4 label the two versions of each minimum wage proposal "(Y)" and "(N)", so the "B" in the parenthesis is also an "N".

2. The share of static minimum wage simulations picking the best proposal

> Treating observed success rates as the truth in simulations of the minimum wage experiment, we picked the best proposal 47% of the time in adaptive experiments and **37%** of the time in static experiments. (p. 837)

Corrected: Treating observed success rates as the truth in simulations of the minimum wage experiment, we picked the best proposal 47% of the time in adaptive experiments and **36.5%** of the time in static experiments.

The deposited simulations pick the best minimum wage proposal in 36.46 per cent of 10,000 static runs, which is 36 per cent at the whole-percentage precision the sentence uses. Rounding to one decimal first gives 36.5, and rounding that again gives 37. The other three shares in the same passage round correctly.

3. The two randomization inference p-values are transposed

> Using randomization inference, we reject the null of no difference across treatments (p = **0.028**). (p. 840)

Corrected: Using randomization inference, we reject the null of no difference across treatments (p = **0.015**).

> Using randomization inference, we again reject the null (p = **0.013**). (p. 841)

Corrected: Using randomization inference, we again reject the null (p = **0.021**).

The first sentence is about the Republican trial and the second about the Democratic one. Run on the deposited data, the deposit's own script returns the published pair with the parties exchanged: 0.013 for the Republicans and 0.028 for the Democrats. The observed F statistics settle which way round it belongs: the Republican F is 2.31 against the Democratic 1.93, and the larger F takes the smaller p-value, so the smaller of the two p-values is the Republicans'. Both nulls are still rejected at conventional levels and no conclusion changes; only the two numbers swap places.

The corrected values are this reproduction's own, from a seeded run of the deposited procedure at 1,000 iterations. They differ in the third decimal from the deposit's unseeded run, which is the sampling variation the procedure carries.

4. The independence that warrants the Horvitz-Thompson factorisation names the realized outcome

> And because $\boldsymbol{Y_i} \perp\!\!\!\perp K_i \mid S_i$, (app. p. 8)

Corrected: And because $\boldsymbol{Y_i(k)} \perp\!\!\!\perp K_i \mid S_i$,

The realized outcome is $Y_i = \sum_k \mathbf{1}\{K_i = k\} Y_i(k)$, a deterministic function of the assignment and the potential outcomes, so it is never independent of K_i given S_i once treatment has an effect. The condition the proof needs is independence of the *potential* outcome, $Y_i(k) \perp\!\!\!\perp K_i \mid S_i$, which is what B.2 assumes two paragraphs earlier and what B.3 assumes again. The expression being factored on this line already carries $Y_i(k)$ rather than Y_i, so the step is valid and only its stated warrant is wrong. No other line of B.2 is affected and the unbiasedness result stands.

5. The variance of the control-augmented estimator drops the covariance between its two terms

> $\operatorname{Var}\left[\hat{\tau}_k^{HT} \mid S_i\right] = \frac{1}{N^2}\sum_{i=1}^{N} \operatorname{Var}\left[Y_i(k)\frac{\mathbf{1}\{K_i=k\}}{\pi_i(k;S_i)} \,\middle|\, S_i\right] + \frac{1}{N^2}\sum_{i=1}^{N} \operatorname{Var}\left[Y_i(0)\frac{\mathbf{1}\{K_i=0\}}{\pi_i(0;S_i)} \,\middle|\, S_i\right]$ (app. p. 10)

Corrected: $\operatorname{Var}\left[\hat{\tau}_k^{HT} \mid S_i\right] = \frac{1}{N^2}\sum_{i=1}^{N} \operatorname{Var}\left[Y_i(k)\frac{\mathbf{1}\{K_i=k\}}{\pi_i(k;S_i)} \,\middle|\, S_i\right] + \frac{1}{N^2}\sum_{i=1}^{N} \operatorname{Var}\left[Y_i(0)\frac{\mathbf{1}\{K_i=0\}}{\pi_i(0;S_i)} \,\middle|\, S_i\right] + \boldsymbol{\frac{2}{N^2}\sum_{i=1}^{N} \Pr[Y_i(k)=1]\Pr[Y_i(0)=1]}$

$\hat{\tau}_k^{HT}$ is a difference of two Horvitz-Thompson means, so its variance carries $-2\,\mathrm{Cov}$ and the display is a sum of two variances with nothing in that place. The covariance does not vanish. Within a unit the two indicators are mutually exclusive, so the product of the two terms is identically zero and their conditional covariance is $-\Pr[Y_i(k)=1]\Pr[Y_i(0)=1]$, which enters the variance with the opposite sign. The printed expression therefore understates the variance by the term shown.

Neither conclusion drawn from the display moves. The missing term is free of the assignment probabilities, so it leaves the minimisation untouched: minimising $\Pr[Y_i(k)=1]/\pi_i(k;S_i) + \Pr[Y_i(0)=1]/\pi_i(0;S_i)$ at fixed total probability still gives probabilities proportional to the square roots of the success rates, and the comparison against the static design at $1/(K+1)$ still favours the adaptive design for $K > 1$. Entry 6 corrects the same display's conditioning and the two compose: the term added here is the one that survives into the unconditional variance entry 6 states.

6. The variance display conditions on one unit's history while summing over all of them

> **Conditional on history, the** variance can be expressed as, $\boldsymbol{\operatorname{Var}\left[\hat{\tau}_k^{HT} \mid S_i\right]} = \frac{1}{N^2}\sum_{i=1}^{N} \operatorname{Var}\left[\,\cdot \mid S_i\right] + \cdots$ (app. p. 10)

Corrected: **Because the summands are a martingale difference sequence with respect to the history, the** variance can be expressed as, $\boldsymbol{\operatorname{Var}\left[\hat{\tau}_k^{HT}\right]} = \frac{1}{N^2}\sum_{i=1}^{N} \mathrm{E}\operatorname{Var}\left[\,\cdot \mid S_i\right] + \cdots$

The index i is free on the left of the printed equality and bound by the sum on the right. Neither reading of the conditioning supports the display. Conditioning on one unit's history cannot license summing per-unit variances, because under adaptive assignment the units are dependent;

conditioning on the whole history leaves only the last unit random, and the sum collapses to a single term.

The sum-of-variances structure is nonetheless right, for the unconditional variance. Under the constant means the section assumes, E of each summand given the history is the constant τ_k, so the centred summands are a martingale difference sequence and every cross-unit covariance is zero. What the section needs is that argument rather than a conditioning bar. The right-hand side is abbreviated above because entry 5 is what changes it; the two entries mark disjoint spans of one display and are applied together.

7. Arm two's posterior density carries arm one's parameters

$$\int_{\theta_2=0}^{\theta_1} c_{2,t}\,\theta_2^{\alpha_{1,t}-1}(1-\theta_2)^{\beta_{1,t}-1}\,d\theta_2 \text{ (app. p. 14)}$$

Corrected: $\int_{\theta_2=0}^{\theta_1} c_{2,t}\,\theta_2^{\alpha_{2,t}-1}(1-\theta_2)^{\beta_{2,t}-1}\,d\theta_2$

The inner integral is arm two's posterior density. Its normalising constant $c_{2,t}$ is indexed correctly and both exponents are not: they carry arm one's parameters.

The line immediately below, which substitutes $t = 1$, uses arm two's parameters correctly, $\Gamma(3)/\Gamma(1)\Gamma(2)$ and $\theta_2^0(1-\theta_2)^1$, which is the Beta$(1, 2)$ posterior the preceding paragraph derives for arm two. So the defect is confined to the general display, and the 5/6 the calculation evaluates to is right.

8. The expected best-arm sample mean is offered as a bias that adaptive assignment introduces

> If we condition on the best observed arm, however, the expected sample mean of the best arm is 0.708, although the true parameter value for either arm is 0.5. (app. p. 16)

The number is right. Enumerating the example's sixteen paths returns 0.708, and the two-period enumeration in Table C.1 reproduces in both of its columns. What the number cannot carry is the weight the section puts on it. C.2 exists to show that "this preferential assignment probability conditional on observed success or failure introduces a source of bias into estimation", and both figures in the enumeration are offered under that heading.

The first one earns it. $\mathrm{E}[\hat{\theta}_k] = 0.458$ against a truth of 0.5 is exactly the bias the section is about, and a static 50-50 design run through the same enumeration returns 0.500, unbiased. The second one does not. The maximum of two noisy estimates exceeds their common mean under any design whatever, adaptive or not, and the same static 50-50 design returns 0.750 here: **higher** than the 0.708 the adaptive design produces. Read as the section invites, the figure points the wrong way, because more of the second period's probability goes to the arm that is currently ahead, and that arm's mean is the one a failure pulls down.

Nothing is corrected on the page. The sentence states a true number and the defect is in what a reader is invited to conclude from it, which is not a thing an erratum can fix by replacing a token; rewriting the section's framing would be editing the authors' argument rather than correcting their record. The entry stands as a note so that the finding is on the record beside the sentence it

concerns. Both comparisons above are computed at render time from the setup C.2 itself states, and the computation is held to reproducing the appendix's own 0.458 and 0.708 before its counterfactual is reported.

9. The arm carrying the 4.1-point Republican effect

> In this particular case, the adaptive design with the smaller standard error allows us to declare the 4.1-point average effect of the **accuracy** treatment statistically significant (p = 0.014), whereas under that static design the effect would not have attained statistical significance (p = 0.152). (p. 841)

Corrected: In this particular case, the adaptive design with the smaller standard error allows us to declare the 4.1-point average effect of the **Lottery** treatment statistically significant (p = 0.014), whereas under that static design the effect would not have attained statistical significance (p = 0.152).

Figure 6 gives the Republican Lottery arm an effect of 4.1 percentage points with a p-value of 0.014. The Accuracy arm's Republican estimate is -5.3 percentage points, in the opposite direction, with a p-value of 0.314. The two preceding sentences and Figure 5 both identify the Lottery as the arm in question. The static-design p-value of 0.152 is left as printed: it comes from a bootstrap the deposited materials do not seed reproducibly, and the reproduction cannot recompute it.

10. "The product of two binomials is also a binomial"

> The $Y_i(k)$ and $\mathbf{1}\{K_i = k\}$ terms are independent **binomials**, and the product of two **binomials** is also a **binomial**, and so we express the variance of their product accordingly. (app. p. 10)

Corrected: The $Y_i(k)$ and $\mathbf{1}\{K_i = k\}$ terms are independent **Bernoullis**, and the product of two **Bernoullis** is also a **Bernoulli**, and so we express the variance of their product accordingly.

As a claim about the binomial distribution the sentence is false: the product of two Binomial(n, p) variables with $n > 1$ is not binomial. It holds for the Bernoulli case, which is the only one used here and the one the section's constant-means assumption sets up, and the display that follows is correct. Writing $p = \Pr[Y_i(k) = 1 \mid S_i]$ and $q = \Pr[K_i = k \mid S_i]$, the product is Bernoulli(pq) with variance $pq(1 - pq)$, which is what is printed. Only the word is wrong.

11. "ams" for "arms"

> They show regret as a function of the expected sum of assignments to suboptimal arms, multiplied by the difference between the best arm mean and the mean under **ams** $k \neq k^*$. (app. p. 11)

Corrected: They show regret as a function of the expected sum of assignments to suboptimal arms, multiplied by the difference between the best arm mean and the mean under **arms** $k \neq k^*$.

A dropped letter. The sentence is otherwise right, and so is the claim it leads to: the expected number of plays of suboptimal arms under Thompson sampling grows as $O(\ln N)$ against $O(N)$, that being $N/(K+1)$, under a random design.

12. Reversed subscripts on the control condition's period-two sampling probability

> Suppose that in period two based on the sampling probabilities $(\tilde{p}_{T1,2},\ \tilde{p}_{T2,2},\ \tilde{p}_{2,C})$, from our period size of nine, we assign six observations to arm T1, one to arm T2, and two to the control. (app. p. 19)

Corrected: Suppose that in period two based on the sampling probabilities $(\tilde{p}_{T1,2},\ \tilde{p}_{T2,2},\ \tilde{p}_{C,2})$, from our period size of nine, we assign six observations to arm T1, one to arm T2, and two to the control.

The control condition's probability is set four lines above as $\tilde{p}_{C,2}$ and is written that way everywhere else in the section, arm first and period second, as it is for both treatment arms in the same tuple. Only this one is reversed. Every number in C.3 is right: the posteriors, the 0.929 and 0.713, the cumulative sample difference of four, and both periods' probabilities, which sum to one.

13. Pallmann et al. (2018) skips its ninth author and runs two words of its title together

> Pallmann, Philip, Alun W. Bedding, Babak Choodari-Oskooei, Munyaradzi Dimairo, Laura Flight, Lisa V. Hampson, Jane Holmes, Adrian P. Mander, Matthew R. Sydes, Sofía S. Villar, et al. 2018. "Adaptive Designs in Clinical Trials: Why **UseThem**, and How to Run and Report Them." BMC Medicine 16(1): 29. (p. 844)

Corrected: Pallmann, Philip, Alun W. Bedding, Babak Choodari-Oskooei, Munyaradzi Dimairo, Laura Flight, Lisa V. Hampson, Jane Holmes, Adrian P. Mander, **Lang'o Odondi**, Matthew R. Sydes, Sofía S. Villar, et al. 2018. "Adaptive Designs in Clinical Trials: Why **Use Them**, and How to Run and Report Them." BMC Medicine 16(1): 29.

Two defects in one entry. The names it prints ninth and tenth are the article's tenth and eleventh authors: Lang'o Odondi is skipped, so the entry is not the first ten authors of anything. Crossref, doi 10.1186/s12916-018-1017-7: Pallmann, Bedding, Choodari-Oskooei, Dimairo, Flight, Hampson, Holmes, Mander, Odondi, Sydes, Villar, Wason, Weir, Wheeler, Yap, Jaki. The insertion is marked on the corrected side only, since no span of the published list is wrong in itself. The title's "Use Them" is set without its space, which the rendered page confirms rather than the text extraction.

14. Trippa et al. (2012) prints a first page in place of a page range

> Journal of Clinical Oncology 30(26): **3258**. (p. 844)

Corrected: Journal of Clinical Oncology 30(26): **3258–63**.

Crossref, doi 10.1200/JCO.2011.39.8420, gives the article's pages as 3258 to 3263.

15. Villar, Bowden, and Wason (2015) prints a first page in place of a page range

Statistical Science 30(2): **199**. (p. 844)

Corrected: Statistical Science 30(2): **199–215**.

Crossref, doi 10.1214/14-STS504, gives the article's pages as 199 to 215.